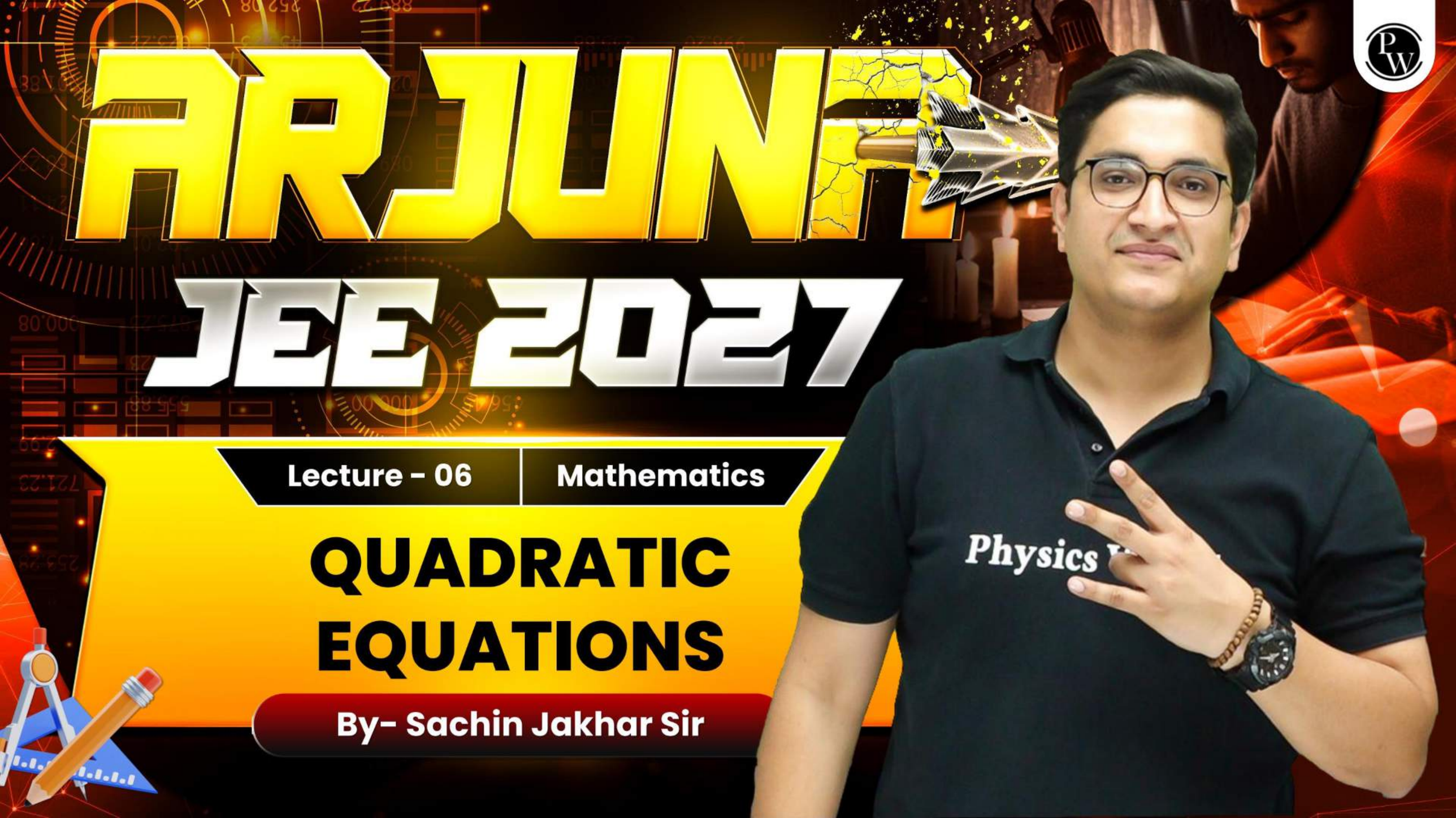
ARJUNA JEE 2027

Lecture - 06 | Mathematics

QUADRATIC EQUATIONS

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Topics *to be covered*

A Types of Roots

B Graph of Quadratic

C Range of Quadratic

D Question Practice



Question Practice

Question [JEE Mains-2023]

Let α, β be the roots of the quadratic equation $x^2 + \sqrt{6}x + 3 = 0$. Then

$$\frac{\alpha^{23} + \beta^{23} + \alpha^{14} + \beta^{14}}{\alpha^{15} + \beta^{15} + \alpha^{10} + \beta^{10}}$$ is equal to

(A) 729

(B) 72

(C) 81

(D) 9

$\alpha^6 + \beta^6 = \alpha^4 \cdot \alpha^2 + \beta^4 \cdot \beta^2$

$= \alpha^4(\alpha^2 + \beta^2) = 0$

$\# \alpha + \beta = -\sqrt{6}$

$\# \alpha\beta = 3$

SBS

$\alpha^2 + \beta^2 + 2(\alpha\beta) = 6$

$\# \alpha^2 + \beta^2 = 0$

$\alpha^2 = -\beta^2$

SBS

$\alpha^4 + \beta^4 + 2\alpha^2\beta^2 = 0$

$\alpha^4 + \beta^4 = -2(\alpha\beta)^2$

$\alpha^4 + \alpha^4 = -18$

$2\alpha^4 = -18$

$\alpha^4 = -9 = \beta^4$

SBS

$\alpha^4 = \beta^4$

$\alpha^8 = \beta^8$

$\alpha^8 = 81$

$\frac{81(\alpha^{15} + \beta^{15})}{\alpha^{15} + \beta^{15}} = 81$

$\alpha^{14} + \beta^{14} = \alpha^4 \cdot \alpha^{10} + \beta^4 \beta^{10}$

$= \alpha^4(\alpha^{10} + \beta^{10}) = 0$

$\alpha^{10} + \beta^{10} = \alpha^4 \cdot \alpha^6 + \beta^4 \cdot \beta^6 = \alpha^4(\alpha^6 + \beta^6)$

$\alpha^{23} + \beta^{23} = \alpha^{15} \cdot \alpha^8 + \beta^{15} \cdot \beta^8$

$= \alpha^8(\alpha^{15} + \beta^{15}) = 81(\alpha^{15} + \beta^{15})$

Complex No.

Question [JEE Mains-2020]

8

79%

The least positive value of 'a' for which the question, $2x^2 + (a-10)x + \frac{33}{2} = 2a$ has real roots is

$$2x^2 + (a-10)x + \frac{33}{2} - 2a = 0$$

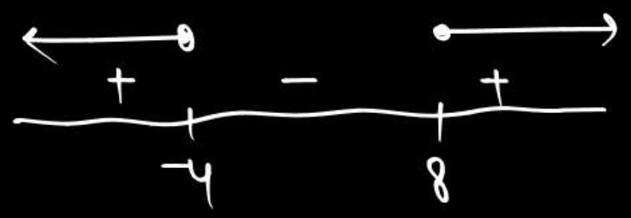
$$D \geq 0$$

$$(a-10)^2 - 4 \times 2 \times \left(\frac{33}{2} - 2a\right) \geq 0$$

$$a^2 + 100 - 20a - 4(33 - 4a)$$

$$a^2 - 4a - 32 \geq 0$$

$$(a-8)(a+4) \geq 0$$



$$a \in (-\infty, -4] \cup [8, \infty)$$

Question

For what value of p, $(4-p)x^2+(2p+4)x+(8p+1)=0$ is a perfect square.

Ans: 0,3

$D=0$

Challenger Question

Let $a, b \in R, a \neq 0$ be such that the equation, $ax^2 - 2bx + 5 = 0$ has a repeated root α, which is also a root of the equation, $x^2 - 2bx - 10 = 0$. If β is the other root of this equation, then $\alpha^2 + \beta^2$ is equal to:

A. 25

B. 28

C. 26

D. 24

$ax^2 - 2bx + 5 = 0$ — roots α, α

$\alpha + \alpha = \frac{2b}{a}$ → $2\alpha = \frac{2b}{a}$

$\alpha\alpha = \frac{5}{a}$

$\alpha^2 = \frac{5}{\left(\frac{1}{4}\right)} = 20$

$D = 0$

$4b^2 - 4 \times a \times 5 = 0$

$\# \; b^2 = 5a$

$x^2 - 2bx - 10 = 0$ — roots α, β

$\alpha + \beta = 2b$

$\alpha\beta = -10$ → $\frac{b}{a} \times \beta = -10$

$\# \; \beta = -\frac{10a}{b}$

SBS.

$\alpha^2\beta^2 = 100$

$20\beta^2 = 100$

$\beta^2 = 5$

$\frac{b}{a} - \frac{10a}{b} = 2b$

$b^2 - 10a^2 = 2ab^2$

$5a - 10a^2 = 2a(5a)$

$5a - 10a^2 = 10a^2$

$5a = 20a^2$

$a = 0, \frac{1}{4}$ ✓

$\alpha^2 + \beta^2$ → 20 + 5

Types of Roots: Sachin Sir Special Summary (S^4):

$$ax^2 + bx + c = 0 \quad (a, b, c \in R), \ (a \neq 0)$$

- $D = 0$ (Roots are real & equal)
- $D > 0$ (Roots are real & distinct)
  - $a, b, c \in Q$, $D \equiv$ Perfect square (Roots are Rational) ✓
    - $a = 1, b \ \& \ c \in I$, $D \equiv$ Perfect square (Roots are Integral)
  - $a, b, c \in Q$ ✓, $D \equiv$ Not a perfect square ✓ (Roots are Irrational)
    - $p + \sqrt{q}, p - \sqrt{q}$ pair.
- $D < 0$ (Roots imaginary)
  - $p + iq, p - iq$ Conjugate pair.

$x^2 - 5x + 6 = 0$ ⟨ 2, 3

$a = 1$, $b = -5$, $c = 6$

$D = 25 - 4 \times 6$

$D = 1$

$$x = \frac{-b \pm \sqrt{D}}{2a} \xrightarrow{a=1} = \frac{-b \pm \sqrt{b^2-4ac}}{2}$$

$x^2 + 2x - 15 = 0$ ⟨ −5, 3

$a = 1$ $b = 2$, $c = -15$

$D = 4 - 4 \times 1 \times (-15) = 64$

b	$b^2 - 4ac$
odd	odd − even = odd → P.S.

$$x = \frac{-\text{odd} \pm \text{odd}}{2a} = \frac{\text{even}}{2a} = \text{Int.}$$

$2a \hookrightarrow 1$

PLEASE CHECK THE FOLLOWING

Ex. : $x^2 - (2 + \sqrt{3})x + \sqrt{12} = 0$ → 2, $\sqrt{3}$

Irr. $2\sqrt{3}$

Ex. : $x^2 - (2 + i)x + 2i = 0$ → i, 2

Ex. : Form Quadratic whose one roots is $2 + \sqrt{3}$.) Incomplete info

Ex. : Form Quadratic with rational coefficient having one root is $2 + \sqrt{3}$.

$2 - \sqrt{3}$.

→ $x^2 - 4x + 1 = 0$

Question [JEE Mains-2021]

$\frac{4}{\sqrt{5}-1}$ is a root of equation

A $x^2 + 2x - 4 = 0$

B $x^2 - 2x + 4 = 0$

C $4x^2 + 2x - 1 = 0$

D $x^2 - 2x - 4 = 0$

$$\frac{4}{\sqrt{5}-1} \times \frac{\sqrt{5}+1}{\sqrt{5}+1} = \frac{4(\sqrt{5}+1)}{(\sqrt{5})^2-(1)^2} = \frac{4(\sqrt{5}+1)}{4} = 1+\sqrt{5}$$

$1-\sqrt{5}$

$x^2-(2)x+(-4)=0$ → $1+\sqrt{5}$, $1-\sqrt{5}$

IMPORTANT OBSERVATION

Quadratic : $ax^2 + bx + c = 0$

(i) If $a + b + c = 0$ (Sum of coeff) $\Rightarrow$ Roots of quadratic are $\rightarrow$ $1, \frac{c}{a}$

(ii) If $a - b + c = 0$ $\Rightarrow$ Roots of quadratic are $\rightarrow$ $-1, -\frac{c}{a}$

#

$$7x^2 - 8x + 1 = 0$$

$$7x^2 - 7x - x + 1 = 0$$

$$7x(x-1) - 1(x-1) = 0$$

$$(x-1)(7x-1) = 0$$

$$\boxed{x = 1, \frac{1}{7}}$$

#

$$7x^2 - 8x + 1 = 0$$

$1 \times 7 = 7$, $1 + 7 = 8$

$$\frac{1}{7}$$

#

$$23x^2 - 11x - 12 = 0$$

$1 \times (-12) = -12$, $1 - 12 = -11$

$$\frac{-12}{23}$$

$ax^2 + bx + c = 0$ — roots: 1 and β

$1 \times \beta = \frac{c}{a}$

$1 + \beta = -\frac{b}{a}$

$\beta = -\frac{b}{a} - 1$

$a + b + c = 0$.

$\div a$

$1 + \frac{b}{a} + \frac{c}{a} = 0$.

$\frac{c}{a} = -\frac{b}{a} - 1$

Acidic Question

If $p(q-r)x^2 + q(r-p)x + r(p-q) = 0$ has equal roots prove $\frac{2}{q} = \frac{1}{p} + \frac{1}{r}$

$D = 0$.

M-1

$$(q(r-p))^2 - 4p(q-r)r(p-q) = 0$$

$$q^2(r-p)^2 = 4pr(q-r)(p-q)$$

$D = 0$.

M-2

Task.

Acidic Question

If $p(q-r)x^2 + q(r-p)x + r(p-q) = 0$ has equal roots prove $\frac{2}{q} = \frac{1}{p} + \frac{1}{r}$

1

$\frac{c}{a} = \frac{r(p-q)}{p(q-r)}$

(M-2)

Sum of coeff $= p(q-r) + q(r-p) + r(p-q)$

$= pq - pr + qr - pq + rp - rq$

$= 0$.

$\frac{r(p-q)}{p(q-r)} = 1$.

$rp - rq = pq - pr$

$pqr \div$

$2rp = pq + rq$.

$\frac{2rp}{pqr} = \frac{pq}{pqr} + \frac{rq}{pqr}$.

$\frac{2}{q} = \frac{1}{r} + \frac{1}{p}$

HHPP.

Question

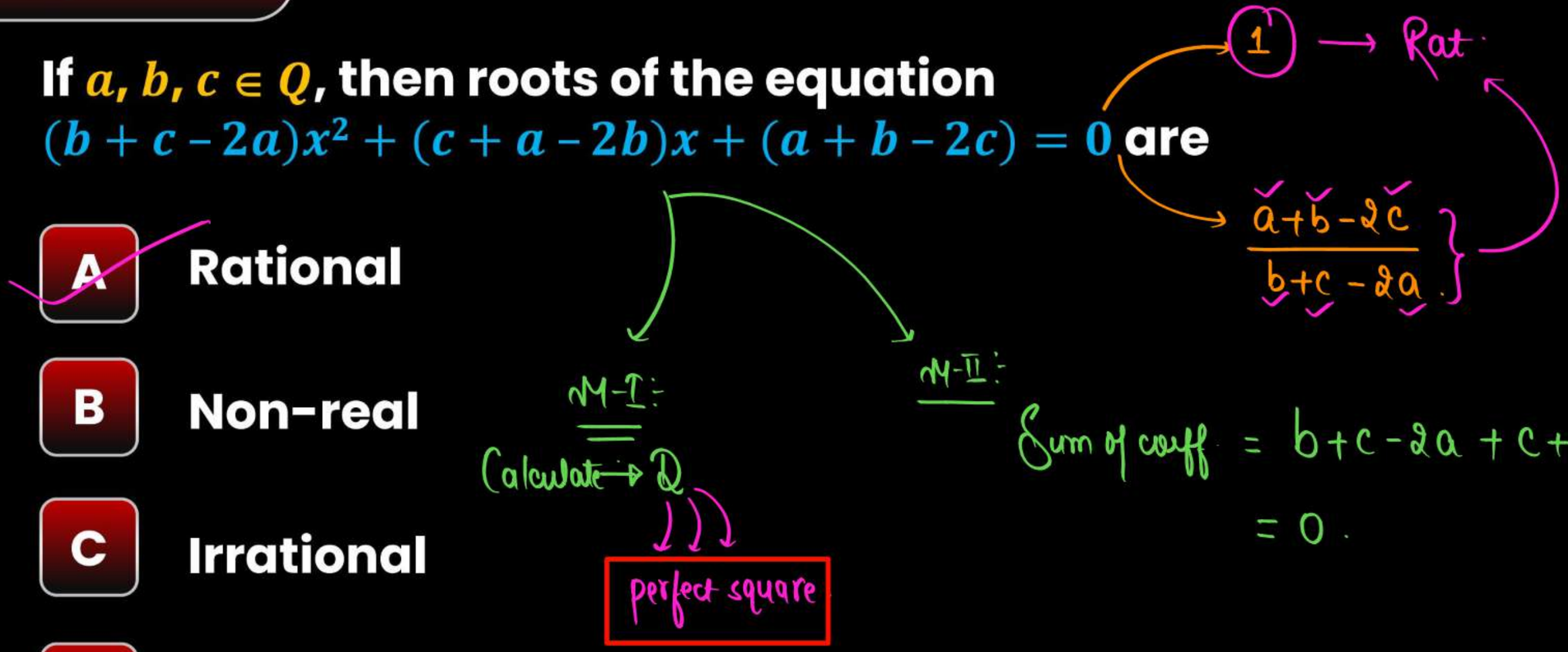
If $a, b, c \in Q$, then roots of the equation $(b+c-2a)x^2+(c+a-2b)x+(a+b-2c)=0$ are

A Rational

B Non-real

C Irrational

D Equal



Challenger Question [JEE Mains-2025]

Consider the equation $x^2 + 4x - n = 0$, where $n \in [20, 100]$ is a natural number. Then the number of all distinct values of n, for which the given equation has integral roots, is equal to

A. 7

B. 8

C. 6

D. 5

M-1

$\mathcal{D}$ = perfect square.

$\mathcal{D} = (4)^2 - 4 \times 1 \times (-n)$

$= 16 + 4n$

$\mathcal{D} = 4(4+n)$

P.S. P.S.

$20 \le n \le 100$

$24 \le n+4 \le 104$

P.S.

$n+4 = 25, 36, 49, 64, 81, 100$

$n = 21, 32, 45, 60, 77, 96.$

M-2

$x^2 + 4x = n$

$x^2 + 4x + 4 = n + 4$

$(x+2)^2 = n+4$

$x + 2 = \pm\sqrt{n+4}$

$\# \quad x = -2 \pm \sqrt{n+4}$

$n+4$ = P.S.

CRITICAL THINKING QUESTION(CTQ)

$a = 12, 8$ fnf.

Find all the integral values of a for which the quadratic equation $(x-a)(x-10)+1=0$ has integral roots.

$x^2 - ax - 10x + 10a + 1 = 0$

$x^2 - (a+10)x + (10a+1) = 0$

$D = (-(a+10))^2 - 4 \times 1 \times (10a+1)$

P.S.

$= a^2 + 100 + 20a - 40a - 4$

$= a^2 - 20a + 96$

$= a^2 - 20a + 100 - 4$

$D = (a-10)^2 - 4 = 0, 1, 4, 9, 16, 25, \ldots\ldots$

(i) $(a-10)^2 - 4 = 0 \Rightarrow (a-10)^2 = 4 \rightarrow a - 10 = \pm 2$

$a = 12, \; a = 8$

(ii) $(a-10)^2 - 4 = 1$

$(a-10)^2 = 5$

$a - 10 = \pm\sqrt{5}$

$a = 10 \pm \sqrt{5}$ ✗

(iii) $(a-10)^2 - 4 = 4$

(iv) $(a-10)^2 - 4 = 9$

$(a-10)^2 = 8$

$a - 10 = \pm 2\sqrt{2}$ ✗

CRITICAL THINKING QUESTION(CTQ)

• $a = 12, 8$ Ans.

Find all the integral values of a for which the quadratic equation $(x-a)(x-10)+1=0$ has integral roots.

M-2

$(x-a)(x-10) = -1$, x & a both are integer.

$I_1 \times I_2 = -1$

• $1 \quad -1$

$x-a=1$ & $x-10=-1$

$x=9$

$9-a=1$

$8=a$

• $-1 \quad 1$

$x-a=-1$ & $x-10=1$

$x=11$

$11-a=-1$

$12=a$

Critical Thinking Question (CTQ)

If both the roots of the quadratic equation $x^2 - (2n + 18)x - n - 11 = 0, n \in I$, are rational, then find the value(s) of n.

Challenger Question

Let a, b, c, d are positive real numbers such that $\frac{a}{b} \neq \frac{c}{d}$, then the roots of the equation $(a^2+b^2)x^2 + 2x(ac+bd) + (c^2+d^2) = 0$ are:

$\frac{a}{c} \neq \frac{b}{d}$

zero. → Q

$a^2x^2 + 2axc + c^2 + b^2x^2 + 2bxd + d^2 = 0$

ϕ (imag. root).

$(ax+c)^2 + (bx+d)^2 = 0$

$\geq 0 \quad \geq 0$

$ax+c=0 \cap bx+d=0$

$x = -\frac{c}{a} \quad x = -\frac{d}{b}$

$+\frac{c}{a} = +\frac{d}{b} \rightarrow$ n.p.

A real and distinct

B real and equal

C imaginary

D nothing can be said

Question [JEE Mains-2022]

If for some $p, q, r \in R$, not all have same sign, one of the roots of the equation $(p^2 + q^2)x^2 - 2q(p + r)x + q^2 + r^2 = 0$ is also a root of the equation $x^2 + 2x - 8 = 0$, then $\frac{q^2 + r^2}{p^2}$ is equal to



DO IT BY YOURSELF (DIBY)

DIBY-27

If roots of the equation $(x-\alpha)(x-\beta)+(x-\gamma)(x-\delta)=0$ are m & n respectively, then roots of the equation $2(x-m)(x-n)-(x-\alpha)(x-\beta)$ Will be

(A) $\frac{\alpha+m+\gamma}{2}, \frac{\beta+n+\delta}{2}$

(B) γ, δ

(C) $\alpha - m, \beta - n$

(D) $m - \alpha + \gamma, n - \beta + \delta$

[Ans. : (B)]

DIBY-28

If the roots of the equation $ax^2+x+b=0$ be real and different, then the roots of the equation $x^2-4\sqrt{ab}x+1=0$ will be

(A) Rational

(B) Irrational

(C) Real

(D) Imaginary

[Ans. : (D)]

DIBY-29

Let α, β are the roots of the equation $x^2+ax+b=0$ such that $\frac{3\alpha-5-\alpha^2}{(\beta-3)}=\beta$, then value of $|a^2-2b+3a|$ is (where, $\beta \neq 3$)

(A) 0 (B) 3 (C) 5 (D) 7

[Ans. : (C)]

DIBY-30

If α is a root of the equation $4x^2 - 2x + 3 = 0$, then its other roots is

(A) $\frac{2}{3} + \alpha$

(B) $\frac{\alpha+1}{2\alpha-1}$

(C) $\frac{3}{2(4\alpha^2+3)}$

(D) $\frac{3\alpha}{4}$

[Ans. : B, C]

DIBY-31

If roots of $2(x-a)(x-b) + c = 0$ are α, β then roots $(x-\alpha)(x-\beta) - \frac{c}{2} = 0$ are

(A) lies between a & b

(B) $-a, -b$

(C) a, b

(D) $2a, 2b$

[Ans. : (C)]

DIBY-32

If the equation in x, $x^2 + 2(1+\lambda)x + (3\lambda^2 + 2) = 0$ has imaginary roots, then

(A) $\lambda = 1$

(B) $\lambda = -\frac{1}{2}$

(C) $\lambda = 0$

(D) $\lambda = -1$

[Ans. : A, B, C, D]

DIBY-33

If the equation $\frac{x^2-bx}{ax-c}=\frac{m-1}{m+1}$ has roots equal in magnitude but opposite in sign, then m is

(A) $\frac{a+b}{a-b}$ (B) $\frac{a-b}{a+b}$ (C) $\frac{b-a}{b+a}$ (D) None of these

[Ans. : (B)]

DIBY-34

Find the value of a for which one root of the equation $x^2+(2a-1)x+a^2+2=0$ is twice as large as the other.

[Ans. : $a=-4$]

DIBY-35

If $x=3+\sqrt{5}$ then the value of $x^4-12x^3+44x^2-48x+17$.

(A) –1 (B) 0 (C) 1 (D) 2

[Ans. (C)]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians